0001 — Pullback Scale-In on the MC CC Setup — 2026-06-24

Series: MC Setup Research Notes · Note 0001 **Confidence:** High — 5 years, full-cost, OOS-tested by year and stress-tested on exit assumptions; the NO-GO holds throughout.

TL;DR: Adding a second contract when price pulls back toward your stop, then targeting the original 1R, **is not worth it on the MC setup.** Across every test — first-principles math, tick-by-tick path analysis on 5 years of data, per-setup and time-of-day and context filters, a full pullback-depth sweep (0.25→1.0R), and an execution stress test — the add never clears the bar it needs to. Its apparent profit is an accounting artifact of marking unresolved trades at the session close; remove that one optimistic assumption and the edge disappears at every depth. **Recommendation: take the clean single-leg trade to 1R and skip the scale-in.** The only durable, real finding is a negative one: pullbacks in the afternoon (after 13:00 CT) are structurally bad and should be avoided regardless.

1. The setup (so this note stands alone)

The MC CC trade. The "MC" indicator fires a breakout signal (one of five subtypes, **CC1–CC5**). You enter at the signal bar's close (**E1**), place a stop a fixed distance away, and target **+1R** (1R = the entry-to-stop distance). On ES, 1 point = $50/contract; a 10-point stop = **1R = $500**.

The scale-in variant we're testing. Instead of one contract, you add a **second contract (leg 2)** if price first pulls back toward your stop by some fraction of R — e.g. a **50% pullback** means leg 2 fills halfway between entry and stop. Both contracts then exit at the original 1R target. The idea: leg 2's cheaper entry boosts the winner.

The three outcomes of any scale-in attempt (a long; shorts mirror), with p = pullback depth:

Outcome	Leg 1	Leg 2	Total	At p=0.50
Clean win — runs to target, never pulls back	+1R	(never added)	**+1R**	+1R
Scale-in win — pulls back, leg 2 fills, both reach 1R	+1R	+(1+p)R	**+(2+p)R**	+2.5R
Scale-in loss — pulls back, leg 2 fills, both stop	−1R	−(1−p)R	**−(2−p)R**	−1.5R

Cost model. $5 round-turn commission + 1 tick ($12.50) slippage **per leg** = $17.50/leg. The scale-in pays it twice.

2. The question

Does the pullback scale-in add money over just taking the clean single-leg trade to 1R — and if so, at what pullback depth, in what conditions?

3. How we tested it (the menu)

1. **First-principles math** — what win rate does the add need to break even? 2. **Tick-by-tick path study** — what happened to all 5,580 signals over 5 years. 3. **Per-setup (CC) breakdown** — does the add help any subtype? 4. **Time-of-day** — does session phase change the picture? 5. **"Tell" hunt** — can any pre-trade signal pick the winners? 6. **Year-by-year validation** — does anything good survive out of sample? 7. **Pullback-depth sweep** — is 50% even the right depth (0.25 → 1.0R)? 8. **Execution stress test** — how much depends on optimistic exit assumptions?

Data: ba_signals_mc.parquet (5,580 signals, 2021-06 → 2026-06), replayed on real continuous ticks day-by-day. 5,444–5,490 have tick coverage. All P&L net of costs.

4. Results

4.1 The math — the add needs a win rate it doesn't have

Per-trade outcomes ($500 R, full cost, 50% PB):

Outcome	Contracts	Net $	Net R
Clean win	1	+$482.50	+0.97R
Scale-in win	2	+$1,215	+2.43R
Scale-in loss	2	−$785	−1.57R

Breakeven win rates by structure (1R target):

Structure	Win	Loss	Breakeven win%
No scale-in (clean 1 contract)	+$482.50	−$517.50	**51.8%**
Scale-in @ 50%, 1R target	+$1,215	−$785	**39.3%**
Scale-in @ 50%, **breakeven exit** (the trap)	+$240	−$785	**76.0%**
Leg 2 alone (the marginal add decision)	+$732.50	−$267.50	**26.8%**

Read: on its own, the second contract only needs to reach 1R on **~27%** of the pullbacks it joins to be worth adding. Hold that 27% — it's the bar for everything below. (The marginal bar widens as the pullback gets shallower: it equals (1−p)/2, i.e. 37.5% at 0.25R, 33.5% at 0.33R, 25% at 0.50R, 17% at 0.66R.)

4.2 The path study — what actually happened (50% pullback, N=5,490)

Outcome	Count	% of all
Clean win (straight to 1R)	1,684	30.7%
Scale-in win (PB → 1R)	538	**9.8%**
Scale-in loss (PB → stop)	1,829	**33.3%**
Clean EOD (no pullback, no target)	789	14.4%
Pullback then EOD (unresolved)	604	11.0%

- **54% of signals pull back** to the 50% level, but only **18% of those then reach** 1R **(538 / 2,971). The biggest single outcome of any signal is** PB → stop (33%) **— the pullback usually precedes a loss, not a bounce.**
- Among resolved pullbacks (1R vs stop), the win rate is **22.7%** — **below the** ~27% the add needs. **On its own bet, the second contract loses.**
- Full population: scale-in net **$306,256** vs single-leg **$268,868** → the add appears to add **+$37,388**. §4.8 shows that's a marking artifact.

4.3 Per-setup (CC) — no subtype rescues it

50% PB, full cost. SI = scale-in, SL = single-leg.

CC	N	PB→1R (% all)	PB-touch%	SI net	SL net	Add value	SI expR	SI PF
CC1	132	5.3%	47.0%	$23,185	$25,108	**−$1,922**	0.061	1.39
CC2	906	10.7%	51.5%	$65,260	$52,170	+$13,090	0.049	1.16
CC3	1,633	11.0%	57.5%	$61,930	$44,612	+$17,318	0.020	1.08
CC4	1,656	10.0%	55.0%	$80,810	$72,685	+$8,125	0.058	1.10
CC5	1,163	7.7%	51.0%	$75,071	$74,292	**+$779**	0.073	1.15

The add is negative for CC1, ~zero for CC5 (the strongest standalone setup), and only meaningfully positive for CC2/CC3 — and §4.8 dissolves even those.

4.4 Time-of-day — the one strong pattern (CME Central Time)

Phase	N	Clean win	PB-touch	PB→1R (of all)	Resolved win%
Open (08:30–11:30)	2,338	33.7%	60.7%	12.4%	23.8%
Mid (11:30–13:00)	1,189	37.1%	57.3%	12.7%	26.7%
Late (13:00–14:45)	1,443	29.0%	54.3%	**6.7%**	17.5%
Close (14:45–15:15)	575	6.4%	15.3%	**0.0%**	—

The whole setup decays through the session. **Open+Mid = 24.7%** resolved (N=1,782) vs **Late+Close = 16.6%** (N=585). At 50% depth the add makes **+$76,924 in Open+Mid** but loses **−$39,535 in Late+Close** — netting the +$37k above. The afternoon eats more than half the morning's add.

4.5 The "tell" hunt — full feature ranking (resolved binary, base 22.7%, N=2,367)

Every cheap pre-trade feature, best-to-worst bucket. <<< = ≥6 pts over base, (low) = ≥6 under. Causal (look-ahead-safe) tags via tag_signals.

Feature	Bucket (win% / N), best → worst
ext_ema (entry vs EMA20, trade dir)	far-pullback-side 30.5/174 <<< · below 27.1/59 · ≈EMA 25.0/76 · above 22.8/149 · chasing-extended 21.8/1909
session phase	Mid 26.7/565 · Open 23.8/1217 · Late 17.5/555 (low)
balance_state	True 25.9/437 · False 22.0/1930
dir_streak (consecutive same-dir)	4+ 25.9/555 · 1 22.6/957 · 2 20.9/512 · 3 20.7/343
AID_DirMatch (Always-In)	against-regime 25.7/451 · with-regime 22.0/1916
prior_ER (prior-day trend)	≥.5 26.2/515 · ≤.2 23.5/729 · .2-.35 21.3/550 · .35-.5 20.6/472
ER_intra_6	.2-.35 25.3/344 · .35-.5 25.2/408 · ≤.2 24.8/302 · ≥.5 20.8/1303
ER_intra_12	≤.2 24.4/746 · .35-.5 23.2/453 · .2-.35 22.1/485 · ≥.5 20.9/670
is_deep_pullback	True 24.3/515 · False 22.3/1852
AID_State	up +1 23.9/1234 · down −1 21.4/1133
dow	Wed 27.3/477 · Mon 24.8/408 · Tue 20.9/487 · Fri 20.7/492 · Thu 20.5/503
SignalType	CC2 24.2/401 · CC4 23.0/718 · CC3 22.9/780 · CC5 21.3/423 · CC1 15.6/45
is_long	long 23.5/1255 · short 21.9/1112
AID_bars since flip	1-3 25.0/541 · 11+ 22.6/1082 · 4-10 21.4/401 · on-flip 21.3/343
ext_vwap	>1σ 23.4/1423 · -1..0 22.3/341 · <-1σ 20.0/110 · 0..1σ 20.0/380
prior_inside_day	False 23.1/2156 · True 19.4/211
prior_adr_ext (prior trend day)	False 23.3/2033 · True 19.5/334

Best 2-feature combos clearing 30% (N≥60): Mid+balance 37.5/72 · Mid+CC4 31.5/178 · CC2+against-regime 31.0/100 · Mid+AID-1-3-bars 30.9/123 · CC5+balance 30.4/79.

All single tells are weak (3–8 pts). Two notes: the streak only helps at **4+** (2–3 are worse than singletons); Always-In points

counter-trend — using it as continuation confirmation hurts (consistent with its prior shelving as a gate). The 30%+ combos sit on N≈70–130 with ±11-pt confidence bands overlapping the base — **overfit, not signal.**

4.6 Year-by-year — only the negative survives

Resolved win% (1R vs stop) by year × phase:

Year	Open	Mid	Late	Open+Mid	Late+Close
2021	23.5 (119)	24.2 (62)	18.3 (71)	**23.8**	17.3
2022	22.7 (247)	21.6 (125)	22.2 (144)	**22.3**	21.9
2023	29.0 (259)	29.5 (105)	12.0 (108)	**29.1**	11.3
2024	19.6 (240)	30.7 (101)	12.0 (100)	**22.9**	11.4
2025	23.7 (249)	31.3 (131)	20.8 (96)	**26.3**	18.9
2026	24.3 (103)	14.6 (41)	19.4 (36)	**21.5**	18.4

Whole-population PB→1R share (of all signals) by year × phase:

Year	Open+Mid	Late+Close
2021	11.8% (363)	5.3% (245)
2022	11.2% (739)	6.9% (465)
2023	15.9% (665)	3.1% (418)
2024	11.9% (656)	3.7% (328)
2025	13.0% (770)	5.2% (381)
2026	9.3% (334)	3.9% (181)

Robust: Late+Close is worse in **6/6 years** (morning pullbacks reach 1R ~2× more often). **Fragile:** "Open+Mid is a positive edge" clears the ~27% bar in only **2 of 6 years** (2023, 2025); 2022 had no phase edge at all. "Mid is the sweet spot" is a 2023–25 artifact (great those years, weak/noise 2021/22/26) — does not generalize.

4.7 Pullback-depth sweep (0.25 → 1.0R) — shallower looks better… but see §4.8

All signals (single-leg baseline net = $268,868), exits marked at session close:

PB depth	Touch%	PB→1R (all)	Add hit% (1R/touch)	Bar to beat	PB→stop%	SI net	Add value
0.25R	71.5%	20.8%	29.1%	37.5%	33.6%	$371,812	**+$102,945**
0.33R	64.6%	16.2%	25.1%	33.5%	33.6%	$329,227	+$60,360
0.50R	54.6%	9.9%	18.1%	25.0%	33.6%	$306,256	+$37,389
0.66R	46.4%	5.2%	11.3%	17.0%	33.6%	$238,047	−$30,821
0.75R	42.6%	3.3%	7.7%	12.5%	33.6%	$230,041	−$38,826
1.00R	33.6%	0.0%	0.0%	0.0%	33.6%	$236,860	−$32,008

Add value ($) by CC × depth:

CC	0.25	0.33	0.50	0.66	0.75	1.00
CC1	+9,546	+7,848	−1,922	−5,228	−4,634	−665
CC2	+36,284	+33,126	+13,090	−5,884	−3,701	−5,320
CC3	+26,568	+15,342	+17,318	−3,373	−859	−10,518
CC4	+6,625	+1,365	+8,125	−17,829	−35,597	−9,678
CC5	+23,922	+2,679	+779	+1,493	+5,964	−5,828

Add hit-rate (PB→1R / touched) by session × depth:

Phase	0.25	0.33	0.50	0.66	0.75
Open+Mid	33.8%	29.3%	21.0%	13.4%	9.3%
Late+Close	19.4%	15.7%	11.1%	5.9%	3.7%

Add value ($) by session × depth:

Phase	0.25	0.33	0.50	0.66	0.75	1.00
Open+Mid	+134,012	+109,518	+76,924	+16,701	−2,047	−23,468
Late+Close	−31,068	−49,159	−39,535	−47,521	−36,779	−8,540

Add value falls monotonically with depth and goes negative beyond ~0.5R. **Red flag:** at every depth the add's hit rate is **below the bar it needs** (29.1% vs 37.5% at 0.25R). The directional bet loses at all depths — the positive numbers must come from somewhere other than reaching the target.

4.8 Execution stress test — the kill shot

The positive add value comes entirely from **unresolved pullbacks marked at the session close** (optimistic: price drifts back and you flatten at 15:15). We re-priced unresolved adds three ways — **close** (mark at session close), **be** (leg 2 exits flat at its entry, neutral), **stop** (count as a full stop, worst):

Add value ($) — ALL signals (N=5,444):

PB	close	breakeven	stop (worst)
0.25R	+102,945	−60,746	−782,705

PB	close	breakeven	stop (worst)
0.33R	+60,360	−127,164	−680,190
0.50R	+37,389	−182,418	−494,086
0.66R	−30,821	−206,005	−344,933
0.75R	−38,826	−185,970	−262,201

Add value ($) — Open+Mid only, the best case (N=3,453):

PB	close	breakeven	stop (worst)
0.25R	+134,012	**+19,347**	−339,575
0.33R	+109,518	−24,575	−322,357
0.50R	+76,924	−91,458	−279,651
0.66R	+16,701	−118,831	−203,712
0.75R	−2,047	−112,103	−157,909

Add value ($) — Late+Close (N=1,991):

PB	close	breakeven	stop (worst)
0.25R	−31,068	−80,092	−443,130
0.33R	−49,159	−102,589	−357,834
0.50R	−39,535	−90,960	−214,435
0.66R	−47,521	−87,174	−141,221
0.75R	−36,779	−73,867	−104,292

Strip the optimism and **every configuration goes negative except one** — 0.25R in the morning, at a trivial **+$19,347 over five years** (~$6/trade). 50% depth swings from +$77k to **−$91k**. Late+Close is negative under every policy and depth.

5. Why it fails

The pullback-to-stop is **more often the start of a loss than a dip to be bought.** On the MC setup, a 50% retrace reaches the original 1R target only ~18% of the time and stops out ~62% of the time — far below the ~27% the second contract needs. So the add is a **negative-expectancy directional bet** that looks profitable only because the backtest marks the leftover unresolved trades at a favorable session close — a drift you can't reliably capture live. Shallower depths don't fix it; they just amplify the same artifact. The math we started with predicted exactly this: a low-hit-rate add against a −1.5R downside can't clear its own bar.

6. Recommendation

Don't scale in. Take the clean single-leg MC trade to 1R and skip the second contract. It needs ~52% wins to break even, the standalone setup is in that range, and you avoid doubling risk into a move that's already against you.

If you scale in anyway: only ever at a **shallow (~0.25R) pullback**, only in the **morning (08:30–13:00 CT)**, and understand you're playing for break-even, not edge.

The one rule worth keeping from all this is negative: avoid the afternoon. Pullbacks after 13:00 CT are structurally bad — they reach target half as often, every year tested. This applies to the base trade too, not just the scale-in.

7. Caveats & open questions

- **EOD marking is the swing variable.** "Close-mark" is optimistic; "breakeven" is neutral; reality is in between. The ranking (shallow >> deep, morning >> afternoon) is robust; the absolute $ is not.
- The sim scores trades **independently** (unlimited concurrent positions) — real capital/overlap constraints would only make the marginal add look worse.
- Path levels use raw signal/stop prices; entry slippage is captured via cost, not geometry.
- with_trend (structural-trend agreement) was broken in this run and excluded — worth a clean re-test, though §4.5 makes a strong tell there unlikely.
- Open follow-up: does avoiding the afternoon improve the **base** single-leg MC trade enough to matter? (Suggested by §4.4 but not isolated here.)

8. Reproduce

Scripts (all headless, day-by-day tick replay):

- scripts/pb_to_1r_path_study.py — path buckets + scale-in vs single-leg P&L (§4.2–4.3)
- scripts/pb_tell_hunt.py — TOD, full tell hunt, year split (§4.4–4.6)
- scripts/pb_level_compare.py — pullback-depth sweep, per-CC & per-session (§4.7)
- scripts/pb_eod_stress.py — EOD policy stress test, all scopes (§4.8)
- scripts/render_note_pdf.py <note.md> — renders any note in this series to PDF

Saved artifacts: docs/living/pb_to_1r_paths.parquet, docs/living/pb_tell_features.parquet, docs/living/pb_level_compare.parquet.